

EPS 522/ENVS 423L Problem Set #5

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Calculations, methods, and assumptions should be spelled out so that your work can be reproduced. All work should be shown to receive full credit.

Upload a word document or PDF of your solutions/answers by midnight on Thursday, October 16 to Canvas.

1 Introduction

The purpose of this problem set is to reinforce your understanding of streamflow and runoff processes.

1.1 Velocity Area Method

Estimate discharge using the velocity-area method for the below measurements.
(15 pts)

Distance from Right Bank (m)	Velocity (m/s)	Depth (m)
0.4	0.23	0.23
0.8	0.17	0.26
1.2	0.19	0.31
1.6	0.24	0.39
2	0.34	0.29
2.4	0.18	0.21
2.8	0.1	0.18
3.3	0.06	0.1

1.2 Endmember Mixing Analysis

Using the provided timeseries data from a CPCRW subcatchment (`cpcrw_catchment.csv`), perform an endmember mixing analysis using the specific conductivity concentrations ($\mu\text{S}/\text{cm}$). Assume the pre-event concentration is 100% and event concentration is 10% of the mean stream conductivity concentration. This will result in some negative discharge values for event flow, assume these are 0. Plot Q_{stream} , $Q_{pre-event}$, and Q_{event} (on the same plot). For the maximum streamflow value, what percent is event vs. pre-event? (20 pts)

1.3 Centroid Lag

Calculate the centroid lag (time difference between precipitation and discharge center of mass timing).

For discharge, you will want to identify the initial rise of the storm hydrograph, subtract out the baseflow value (preceding value) and use the following equation for the center of mass time:

$$t_{qc} = \frac{\sum_{i=1}^n Q_i \cdot t_i}{\sum_{i=1}^n Q_i}$$

For precipitation, use the above equation for the duration of the precipitation storm event. Report the center of mass date for event precipitation and event discharge along with the centroid lag time (in hrs). (20 pts)

1.4 C-Q Analysis

Does specific conductivity appear to be flushed or diluted during the storm? Plot Q and C (on x and y axis respectively) for the storm hydrograph and determine whether it is clockwise or anti-clockwise. What does this suggest about the solute source and storm flowpath length? (20 pts)

1.5 Synthesis

Based on the above three analyses, what can you conclude about the storm runoff-generation mechanisms in this subcatchment? (15 pts)